

Airflow & dbt Documentation

E-Commerce Galaxy Schema Data Pipeline

1. Project Overview

This project implements a modern data engineering pipeline using:

- Apache Airflow for orchestration
- dbt (Data Build Tool) for transformations
- PostgreSQL as the data warehouse
- CSV files as raw data sources

The pipeline processes e-commerce datasets and transforms them into a Galaxy Schema suitable for analytics and reporting.

2. Project Architecture

Data Flow

Raw CSV Files



Airflow Extract Tasks



Staging Layer



dbt Transformations



Dimension Tables



Fact Tables



3. Technologies Used

Technology	Purpose
Apache Airflow	Workflow orchestration
dbt	Data transformation
PostgreSQL	Data warehouse
Pandas	CSV processing
Docker	Containerization
Astro CLI	Airflow local development
Python	ETL logic

4. Dataset Files

The project uses the following CSV datasets:

File Name	Description
customers.csv	Customer information
orders.csv	Orders data
order_items.csv	Products inside each order
products.csv	Product catalog
product_reviews.csv	Customer reviews

5. Airflow Pipeline

DAG Name

etl_galaxy_schema

6. Airflow Responsibilities

Airflow is responsible for:

- Workflow orchestration
 - Task scheduling
 - Dependency management
 - Monitoring pipeline execution
 - Triggering dbt transformations
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7. Airflow ETL Process

Extract Layer

Each Airflow task validates and extracts raw CSV files.

Extract Tasks

Task Name	Purpose
extract_customers	Reads customer dataset
extract_orders	Reads orders dataset
extract_order_items	Reads order items dataset
extract_products	Reads product dataset
extract_product_reviews	Reads reviews dataset

8. Transformation Layer

The transformation process creates:

Dimension Tables

Dimension	Description
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dim_customer	Customer dimension
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dim_product	Product dimension
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dim_date	Date dimension
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Fact Tables

Fact Table	Description
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fact_sales	Sales transactions
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fact_reviews	Product review metrics
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9. Galaxy Schema Design

The warehouse follows a Galaxy Schema architecture.

Advantages

- Supports multiple business processes
 - Optimized for analytics
 - Reduces redundancy
 - Improves reporting performance
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10. Fact Table Relationships

fact_sales

Connected to:

- dim_customer
 - dim_product
 - dim_date
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fact_reviews

Connected to:

- dim_customer
 - dim_product
 - dim_date
-

11. dbt Integration

dbt is used after the Airflow extraction phase.

dbt Responsibilities

- Data cleaning
 - Business transformations
 - Data modeling
 - Testing
 - Documentation
 - Incremental loading
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12. dbt Project Structure

dbt_project/

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|— models/

| |— staging/

| |— marts/

| | |— dimensions/

| | |— facts/

|

|— tests/

|— snapshots/

|— seeds/

└─ macros/

13. dbt Staging Models

The staging layer standardizes raw data.

Examples:

Model	Purpose
stg_customers	Clean customer data
stg_orders	Clean orders
stg_products	Clean products

14. dbt Fact Models

dbt generates analytics-ready fact tables.

Examples:

Model	Purpose
fact_sales	Sales analytics
fact_reviews	Review analytics

15. dbt Dimension Models

Dimension models provide descriptive business entities.

Examples:

Model	Purpose
dim_customer	Customer attributes

Model	Purpose
dim_product	Product details
dim_date	Calendar dimension

16. Airflow + dbt Workflow

Pipeline Sequence

CSV Files



Airflow Extract



PostgreSQL Raw Tables



dbt Staging Models



dbt Mart Models



Fact & Dimension Tables



Power BI Dashboard

17. PostgreSQL Data Warehouse

PostgreSQL stores:

- Raw tables
- Staging tables
- Dimension tables

- Fact tables
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18. Data Quality Checks

Airflow Checks

- File existence validation
- Pipeline execution monitoring

dbt Tests

- Null checks
 - Unique constraints
 - Referential integrity
 - Accepted values tests
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19. Performance Optimization

The pipeline includes:

- Task separation
 - Modular transformations
 - Incremental loading support
 - Surrogate keys
 - Optimized joins
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20. Challenges Faced

Memory Issues (OOM)

Large datasets caused Out Of Memory errors during Airflow execution.

Solution

Instead of passing DataFrames between tasks:

- Tasks return file paths only

- DataFrames are loaded inside transformation tasks

This follows production-grade ETL best practices.

21. Future Improvements

Planned enhancements:

- Full dbt integration
 - Incremental models
 - CI/CD deployment
 - Cloud storage integration
 - Docker Compose orchestration
 - Power BI dashboards
 - Data lineage tracking
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22. Conclusion

This project demonstrates a complete modern data engineering workflow using:

- Airflow orchestration
- dbt transformations
- PostgreSQL warehousing
- Galaxy Schema modeling

The architecture is scalable, modular, and aligned with industry best practices for analytics engineering and business intelligence solutions.